

HGCAL Validation Results

Release Validation Campaigns

CMSSW_16_0_X

CMSSW_16_0_0_pre1_vs_CMSSW_15_1_0_pre6

CloseByParticleGun in CE-E Front section with 120 um

Reconstructed hits calibration;

ReconstructableEnergyOverCPenergy/h EoP CPene 200 calib fraction.png

Std Dev: Old value is zero

ReconstructableEnergyOverCPenergy/h EoP CPene scint calib fraction.png

All values are zero

Simulated hits, digis, reconstructed hits validation

+

Layer clusters

Efficiencies vs layer/globalEfficiencies.png

There are very few values and overlap

Efficiencies vs layer/globalEfficiencies.png

Overlaps decrease as layer numbers increase.

Duplicates vs layer/globalEfficiencies.png

Empty

Duplicates vs layer/globalEfficiencies.png

Empty

FakeRate vs layer/globalEfficiencies.png

Empty

FakeRate vs layer/globalEfficiencies.png

All values are zero

MergeRate vs layer/globalEfficiencies.png

Empty

MergeRate vs layer/globalEfficiencies.png

Empty

Tracksters

+

Tracksters vs SimTracksters from CPs Associatedby Hits

+

Tracksters vs SimTracksters from CPs Associatedby LayerClusters

+

Tracksters vs SimTracksters Associated by Hits

+

Tracksters vs SimTracksters Associated by LayerClusters

+